Solutions – Tutorial 2 –The Instruments of Trade Policy

ECON 308 – International Economic Relations

Fall 2025

Problem 1 – Tariff and Welfare for a Small Country

Given:

$$Q_S = P - 100, \qquad Q_D = 700 - P, \qquad P_W = 200.$$

Small country $\Rightarrow$ world price fixed at \$200; tariff does not affect P_W.

(1) Free-trade equilibrium imports

Under free trade, domestic price equals world price:

$$P^{FT} = P_W = 200.$$

Domestic supply:

$$Q_S^{FT} = 200 - 100 = 100.$$

Domestic demand:

$$Q_D^{FT} = 700 - 200 = 500.$$

Imports:

$$M^{FT} = Q_D^{FT} - Q_S^{FT} = 500 - 100 = 400.$$

$$\boxed{M^{FT} = 400 \text{ tons.}}$$

(2) Equilibrium with a \$20 specific tariff

Tariff $t = 20$ per ton. For a small country, world price remains \$200; domestic consumers pay:

$$P^T = P_W + t = 200 + 20 = 220.$$

Supply at P^T:

$$Q_S^T = 220 - 100 = 120.$$

Demand at P^T:

$$Q_D^T = 700 - 220 = 480.$$

Imports:

$$M^T = Q_D^T - Q_S^T = 480 - 120 = 360.$$

$$\boxed{P^T = 220,\ Q_S^T = 120,\ Q_D^T = 480,\ M^T = 360.}$$

(3) Tariff revenue

Government revenue:

$$\text{Rev} = t \cdot M^T = 20 \times 360 = 7{,}200.$$

$$\boxed{\text{Tariff revenue} = \$7{,}200.}$$

(4) Deadweight losses from production and consumption

The tariff wedges domestic price P^T above P_W by $t = 20$. Quantity changes:

$$\Delta Q_S = Q_S^T - Q_S^{FT} = 120 - 100 = 20,$$

$$\Delta Q_D = Q_D^{FT} - Q_D^T = 500 - 480 = 20.$$

Production distortion (over-production) loss: Triangle at the left side of the import demand diagram with base ΔQ_S and height t:

$$\text{DWL}_{\text{prod}} = \frac{1}{2} \times t \times \Delta Q_S = \frac{1}{2} \times 20 \times 20 = 200.$$

Consumption distortion (under-consumption) loss: Triangle at the right side with base ΔQ_D and height t:

$$\text{DWL}_{\text{cons}} = \frac{1}{2} \times t \times \Delta Q_D = \frac{1}{2} \times 20 \times 20 = 200.$$

$$\boxed{\text{DWL}_{\text{prod}} = 200, \quad \text{DWL}_{\text{cons}} = 200.}$$

(5) Net welfare loss

Total deadweight loss:

$$\text{Total DWL} = \text{DWL}_{\text{prod}} + \text{DWL}_{\text{cons}} = 200 + 200 = 400.$$

In a small country, there is *no terms-of-trade gain*, so the tariff simply transfers surplus from consumers to producers and government, plus this net loss:

$$\boxed{\text{Net welfare loss} = \$400.}$$

Problem 2 – Large Country Tariff and Terms of Trade

Given:

$P^* = 100 + 0.5Q^*$ (foreign export supply), $Q_M^D = 400 - 2P$ (Home import demand).

P^* is the price received by foreign exporters (world price). P is the price paid by Home importers (domestic import price).

(1) Free-trade equilibrium price and imports

Under free trade, there is no tariff, so:

$$P = P^* = P^{FT}, \qquad Q = Q^* = Q_M^D.$$

Use Home import demand:

$$Q = 400 - 2P.$$

Use foreign export supply:

$$P = 100 + 0.5Q.$$

Substitute $Q = 400 - 2P$ into the export supply equation:

$$P = 100 + 0.5(400 - 2P) = 100 + 200 - P = 300 - P.$$

So

$$2P = 300 \Rightarrow P^{FT} = 150.$$

Then

$$Q^{FT} = 400 - 2(150) = 400 - 300 = 100.$$

$$\boxed{P^{FT} = 150, \quad Q^{FT} = 100 \text{ units of imports.}}$$

(2) \$20 tariff: new world price, domestic price, and imports

Let the tariff be $t = 20$. Then

$$P = P^* + t.$$

Equilibrium conditions:

$$Q = Q_M^D = 400 - 2P, \quad P^* = 100 + 0.5Q, \quad P = P^* + 20.$$

Plug $P = P^* + 20$ into Q_M^D:

$$Q = 400 - 2(P^* + 20) = 400 - 2P^* - 40 = 360 - 2P^*.$$

Foreign export supply:

$$P^* = 100 + 0.5Q = 100 + 0.5(360 - 2P^*) = 100 + 180 - P^* = 280 - P^*.$$

So

$$2P^* = 280 \Rightarrow P^* = 140.$$

Then domestic price:

$$P = P^* + t = 140 + 20 = 160.$$

Import quantity:

$$Q = 360 - 2P^* = 360 - 2(140) = 360 - 280 = 80.$$

$$\boxed{P^* = 140, \quad P = 160, \quad Q^T = 80.}$$

Note: World price falls from 150 to 140; Home improves its terms of trade.

(3) Tariff revenue and terms-of-trade gain

Tariff revenue:

$$\text{Rev} = t \cdot Q^T = 20 \times 80 = 1{,}600.$$

$$\boxed{\text{Tariff revenue} = \$1{,}600.}$$

Terms-of-trade (ToT) gain: Before tariff, foreign received $P^{FT} = 150$; after, $P^* = 140$. The drop in the price Home pays to foreigners (in world markets) is

$$\Delta P_{\text{ToT}} = 150 - 140 = 10.$$

Home still imports $Q^T = 80$ units at this lower foreign price, so the ToT gain is:

$$\text{ToT gain} = \Delta P_{\text{ToT}} \cdot Q^T = 10 \times 80 = 800.$$

$$\boxed{\text{Terms-of-trade gain} = \$800.}$$

(4) Welfare comparison: ToT gain vs. deadweight loss

The total welfare effect of the tariff for a large country is:

$$\Delta W = \text{ToT gain} - \text{DWL}.$$

The tariff reduces imports from $Q^{FT} = 100$ to $Q^T = 80$, a change of:

$$\Delta Q = 100 - 80 = 20.$$

Tariff wedge is $t = 20$. In the standard trade diagram (quantity on horizontal, prices on vertical), two small triangles arise at the two edges of the new trade volume. Each has base ΔQ and height t:

$$\text{DWL triangle (each)} = \frac{1}{2} \times t \times \Delta Q = \frac{1}{2} \times 20 \times 20 = 200.$$

Total deadweight loss:

$$\text{DWL}_{\text{total}} = 2 \times 200 = 400.$$

Thus,

$$\Delta W = 800 - 400 = 400 > 0.$$

$$\boxed{\text{Net welfare gain} = \$400 \text{ for Home.}}$$

(5) Why retaliation limits optimal tariffs

In reality, trading partners typically retaliate by imposing their own tariffs or trade barriers. This:

- Reduces demand for Home exports,
- Lowers Home's export prices,
- Shrinks or reverses the terms-of-trade gain.

Hence the net effect can easily become negative for both countries, making aggressive use of "optimal tariffs" politically and economically risky. This is a key reason behind multilateral agreements (e.g., WTO) that discourage such behaviour.

Problem 3 – Effective Rate of Protection

Given:

- World price of car: \$20,000.
- World price of imported engine: \$8,000.
- Tariff on engine: 25%.
- Tariff on finished car: 10%.

Let V_A be domestic value added before protection and V_A' after protection.

(1) Nominal protection rate

The nominal tariff on the finished car is 10%.

$$\boxed{\text{Nominal protection rate on the car} = 10\%.}$$

(2) Effective Rate of Protection (ERP)

Without tariffs (free trade):

$$P_{\text{car}}^{FT} = 20{,}000, \quad P_{\text{engine}}^{FT} = 8{,}000.$$

Domestic value added is:

$$V_A = P_{\text{car}}^{FT} - P_{\text{engine}}^{FT} = 20{,}000 - 8{,}000 = 12{,}000.$$

With tariffs:
Engine tariff 25%:

$$P_{\text{engine}}^{T} = 8{,}000(1 + 0.25) = 10{,}000.$$

Car tariff 10%:

$$P_{\text{car}}^{T} = 20{,}000(1 + 0.10) = 22{,}000.$$

New domestic value added:

$$V_A' = P_{\text{car}}^{T} - P_{\text{engine}}^{T} = 22{,}000 - 10{,}000 = 12{,}000.$$

Compute ERP:

$$\text{ERP} = \frac{V_A' - V_A}{V_A} = \frac{12{,}000 - 12{,}000}{12{,}000} = 0.$$

$$\boxed{\text{Effective rate of protection} = 0\%.}$$

(3) Interpretation

Even though there is a 10% *nominal* tariff on the final car, the protection to *domestic value added* is **zero**:

- The tariff on the input (engine) raises its cost to domestic producers.
- The tariff on the final good raises the car price.
- In this numerical example, these effects cancel out so that domestic value added is unchanged at \$12,000.

Thus the industry is *not* effectively protected, despite the nominal 10% tariff. This illustrates why policymakers must look at the **ERP**, not just nominal tariffs, to understand true protection of domestic production.